

Descriptive Astronomy

Tour of the class / Scales in the Universe/
History / Light / Telescopes / Solar system /
Collisions / Stars / Neutron stars and BH



British-born American astronomer who discovered that stars are made mainly of hydrogen and helium and established that stars could be classified according to their temperatures.

Payne entered the University of Cambridge in 1919.

1

In 1925, Payne was the first person to be awarded a PhD in astronomy at Radcliffe College, at the time the women's branch of Harvard University in Cambridge, Massachusetts

Descriptive Astronomy Fall 25

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Lets visit canvas first !

Relative weight will be assigned as follow:

homework 15%

take home test3 15 %

Test1, Test2, Final 60%

Attendance 5%

popquizzes 10%

There is a popquiz at 11AM Each class. It is online and it will be open for 15 minutes. It is open book and open discussion. The popquiz is based on the previous lecture

Tentative exams:

Test1: Feb 4th, Test2: March 11th, test3: April8th

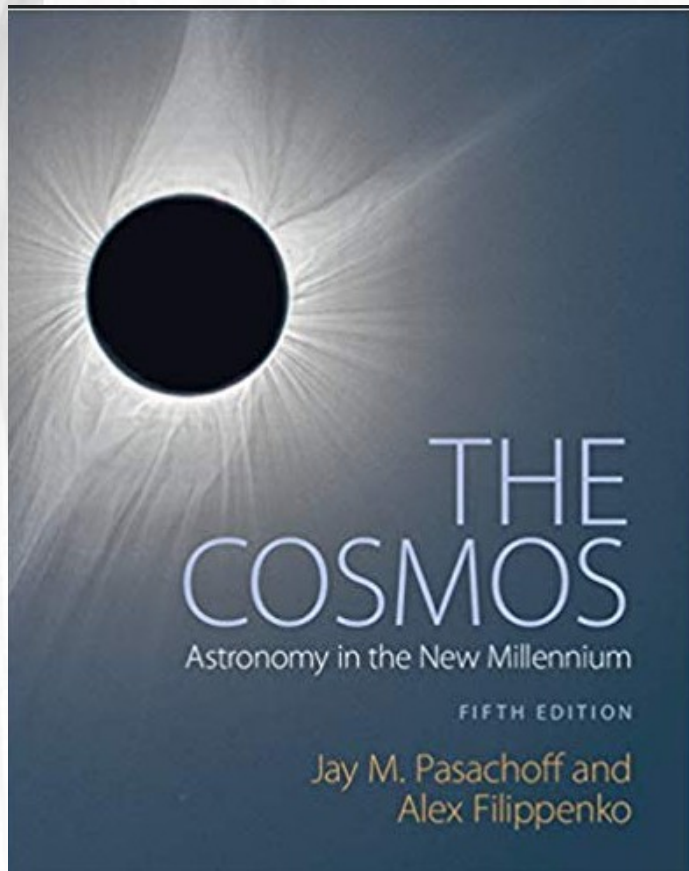
Final TBA



Descriptive Astronomy

All my slides are found in my dropbox. link in Canvas (1st module)
I record the lectures. See my youtube channel.

<https://www.youtube.com/playlist?list=PLYDSr0LB8IcmZc6A-sIXlj6Xmcq95IIIIt>

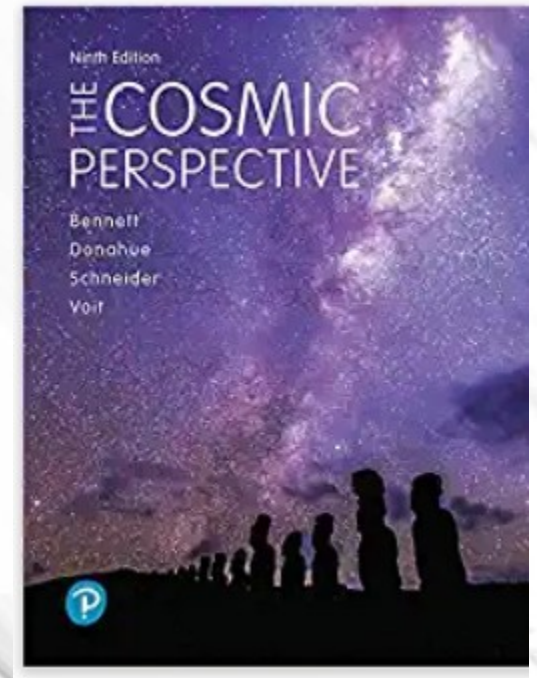


Textbook is optional.

This class is based on that textbook



**The dept
Recommends this
textbook >>>**



Unit 1: Tour of the class. Scale in the Universe.

Time and distance.

Unit 2: Brief history. The shoulder of giants

Unit 3: The sky through binoculars and telescopes.
The celestial sphere. Constellations. More on
telescopes

Unit 4: Overview of the solar system / nebular
hypothesis

Unit 5: Cosmic collisions.

Unit 6: Light the supreme informant. Spectroscopy.

Unit7: Stellar parameters

Unit 8: cosmic ladder, distance in the universe.

Age of objects in the Universe.

Unit 9: Life of stars. Death of stars. Stellar
graveyard.

Unit 10: Black hole and neutron stars.

If time permits: more on cosmology



BEWARE of pseudo Sciences !



PHOTO: ÉTIENNE KLEIN

A photo tweeted by a famous French physicist supposedly of Proxima Centauri by the James Webb Space Telescope was actually a slice of chorizo.

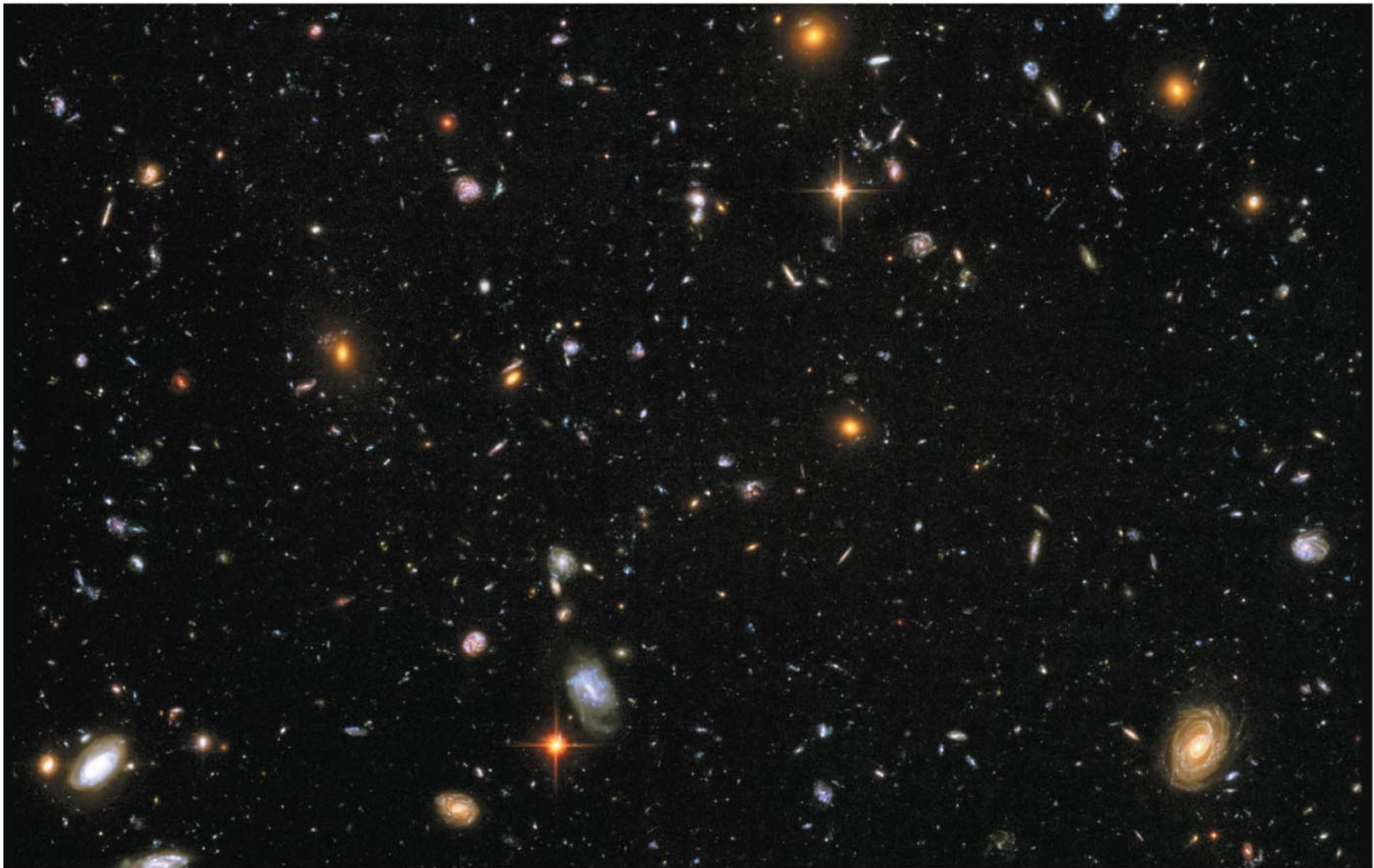


@EtienneKlein · Follow

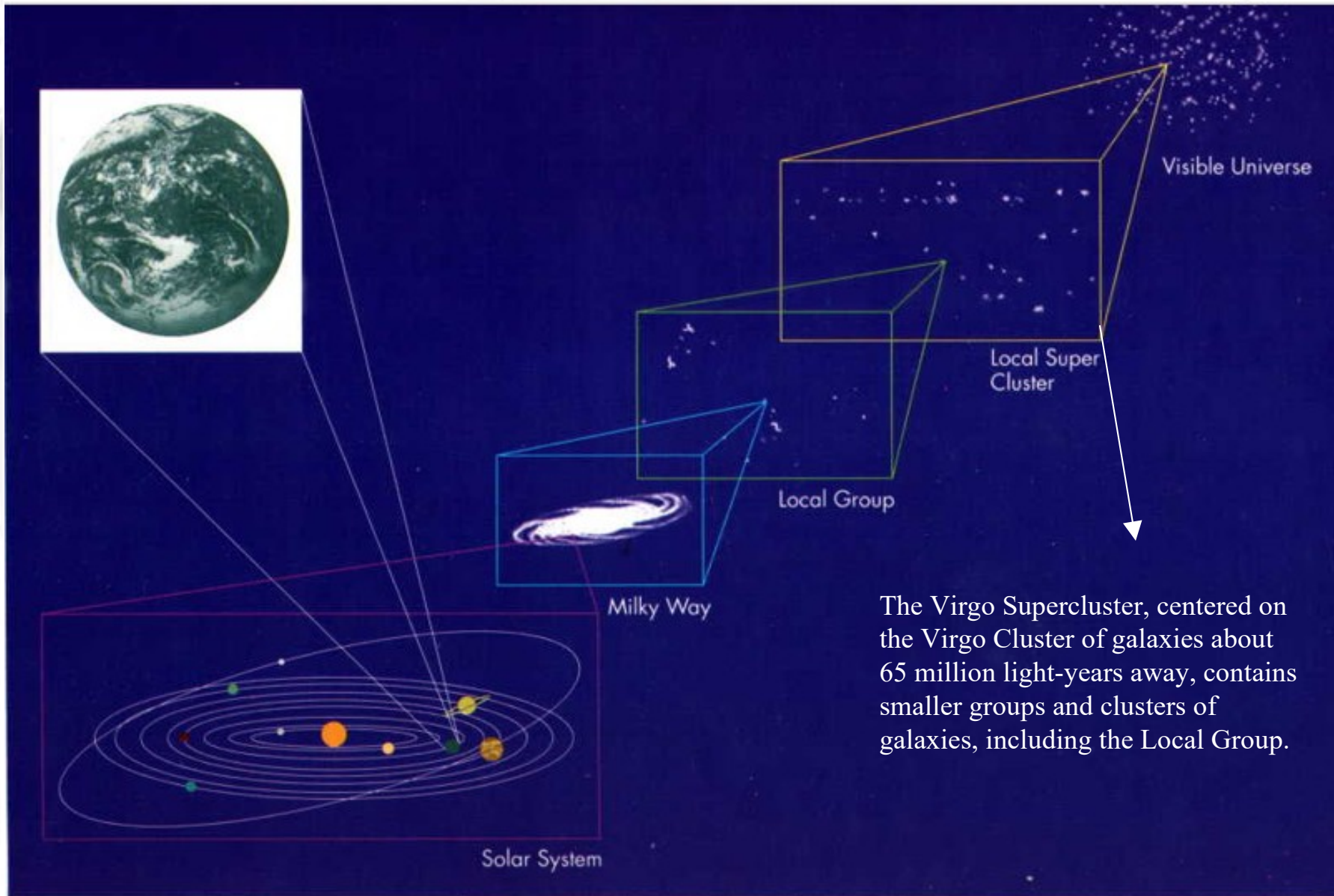
Photo (VRAIE cette fois...) de la galaxie de la Roue du Chariot et de ses galaxies compagnes, prise par le JWST. Située à 500 millions d'années-lumière, elle fut sans doute spirale dans son passé, mais a pris cette étrange allure à la suite d'un furieux carambolage galactique.

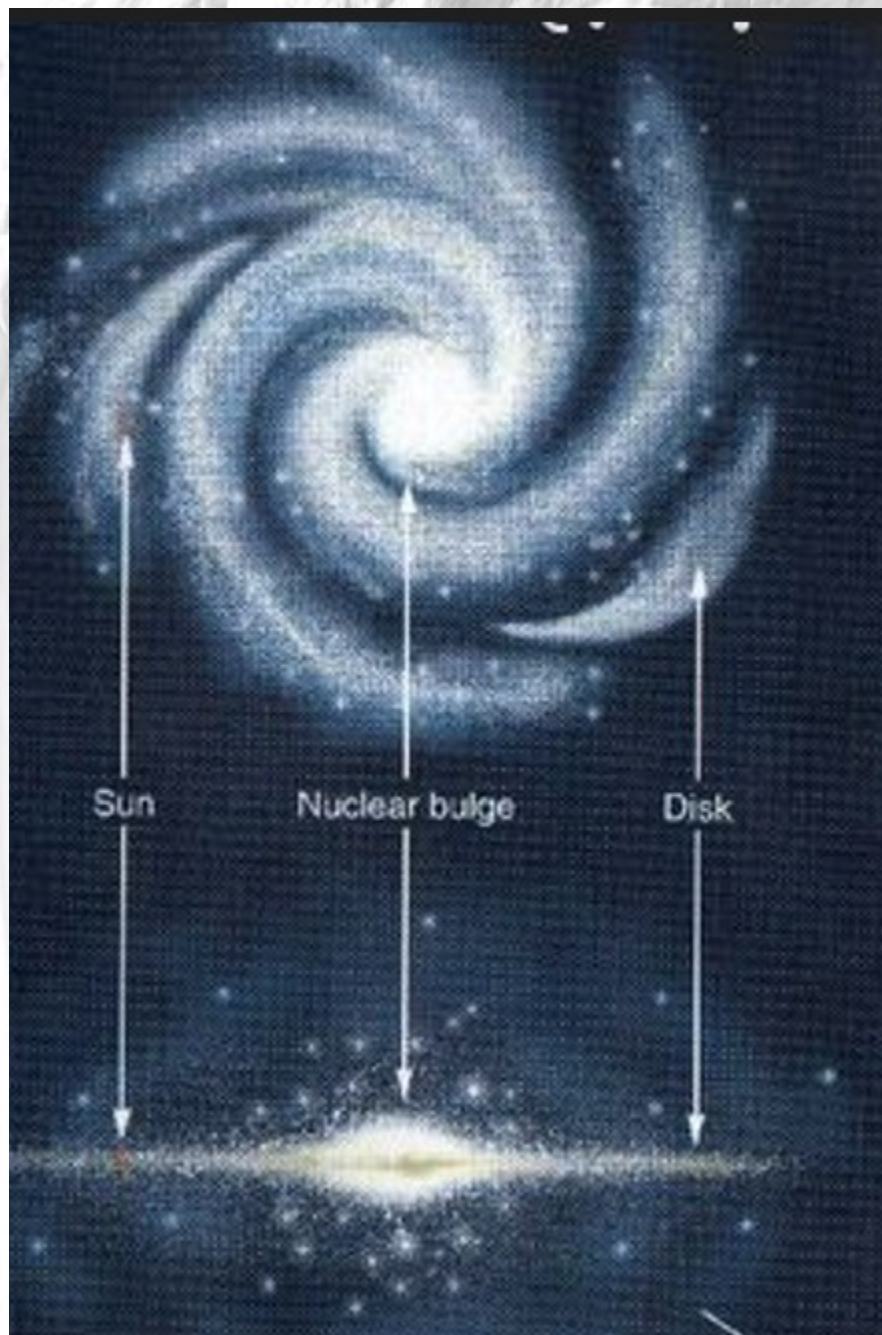


Unit1: Tour of the class

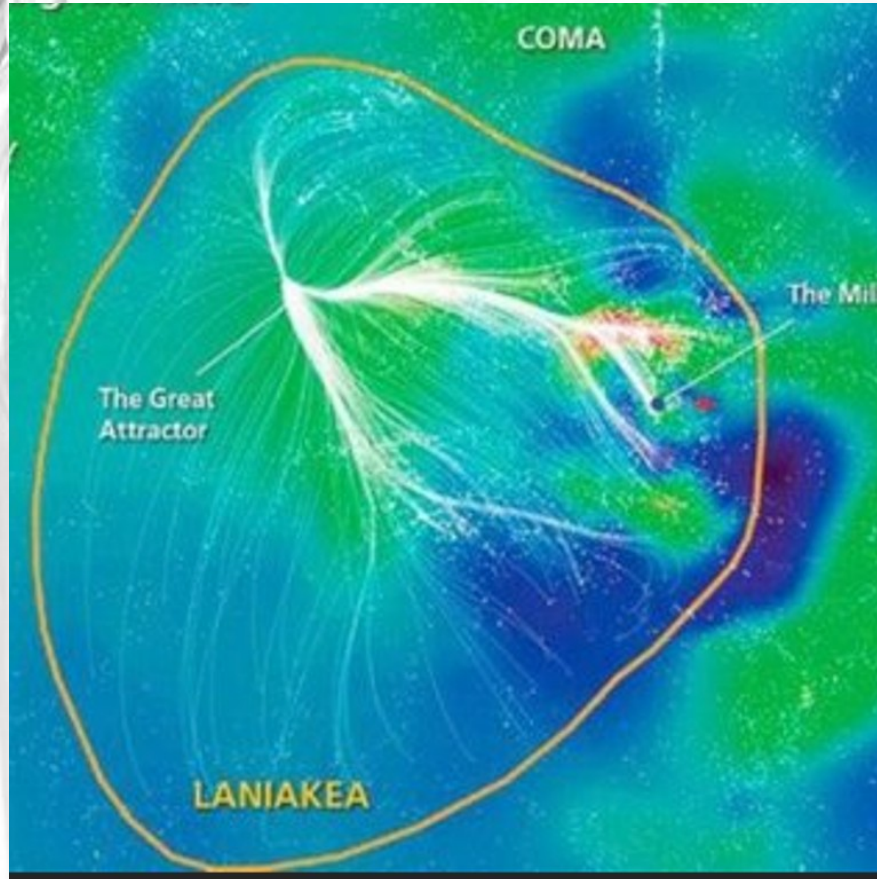


Our place in the Universe. Scale and time.

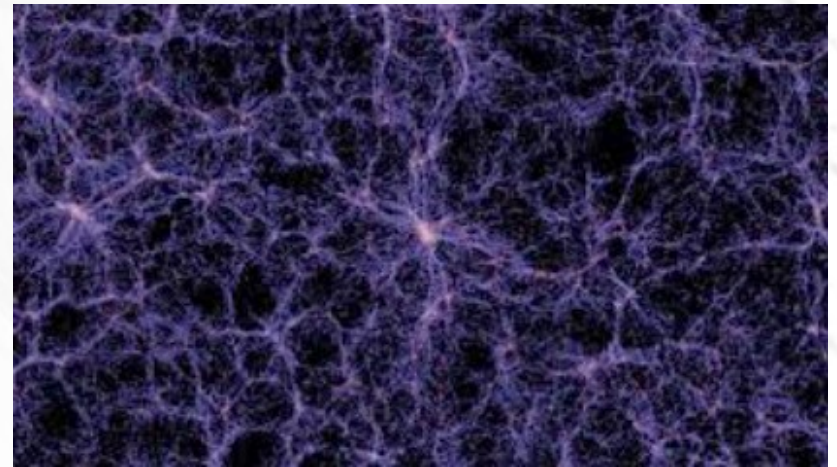




The Laniakea Supercluster was defined in 2014. A team of astronomers, led by R. Brent Tully of the University of Hawai‘i, H  l  ne Courtois of the University of Lyon, Yehuda Hoffman of the Hebrew University of Jerusalem, and Daniel Pomar  de of CEA Universit   Paris-Saclay, published a new way of defining supercluster



Video : Laniakea



<https://www.forbes.com/sites/chadorzel/2019/07/23/indirect-contributions-are-essential-to-physics/>

<https://www.youtube.com/watch?v=rENyyRwxpHo&t=9s>



Andromeda galaxy, 2.5 Million
Light years away.
Belongs to our local group.
Coming toward us.

VIDEO milky way

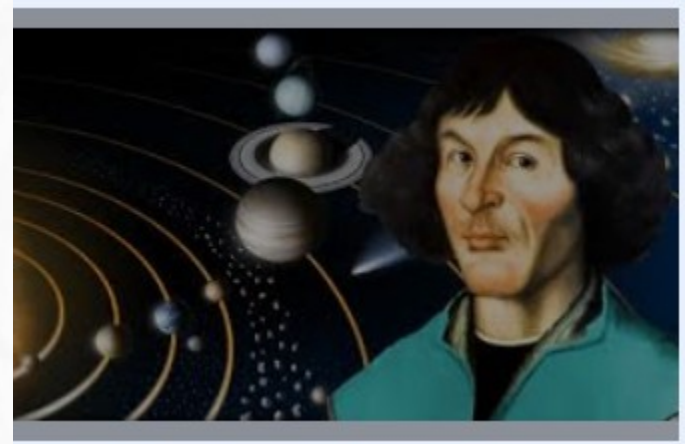
The Andromeda Galaxy, which is far larger than the Milky Way, is hurtling toward us at 68 miles per second. And while that might seem fast, given the distance between these galaxies it will still take 4 billion years for them to collide

Speed of light: 186,000 miles/ 300,000 km/s /9 trillion km in a light year

History



Galileo-Galilei 1564-1642



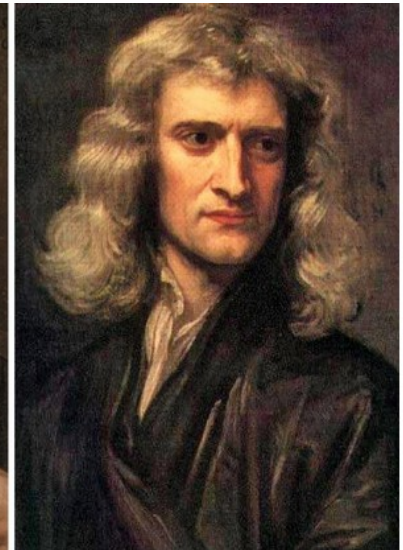
Copernicus 1473-1543



Tycho Brahe
(1546–1601)



Johannes Kepler
(1571–1630)



Isaac Newton
(1643–1727)

constellation



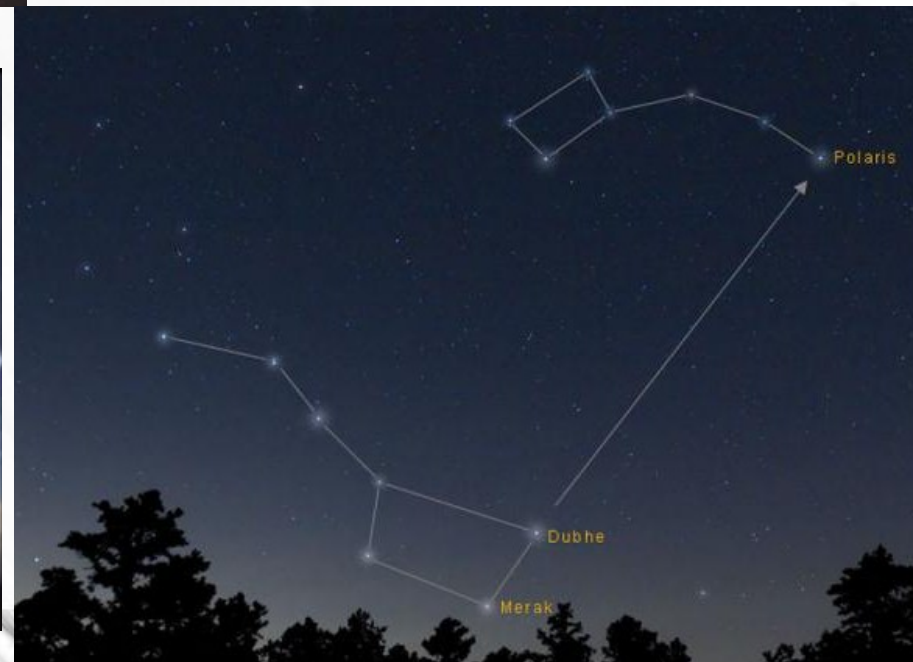
<https://www.skymania.com/wp/northern-hemisphere-sky-chart/>

<https://earthsky.org/tonight/>

<https://www.space.com/16149-night-sky.html>

APP → sky map

Stars rise in the East





<https://www.timeanddate.com/astronomy/night/usa/miami>

Tools of the trades = Telescopes → collect light → great informant

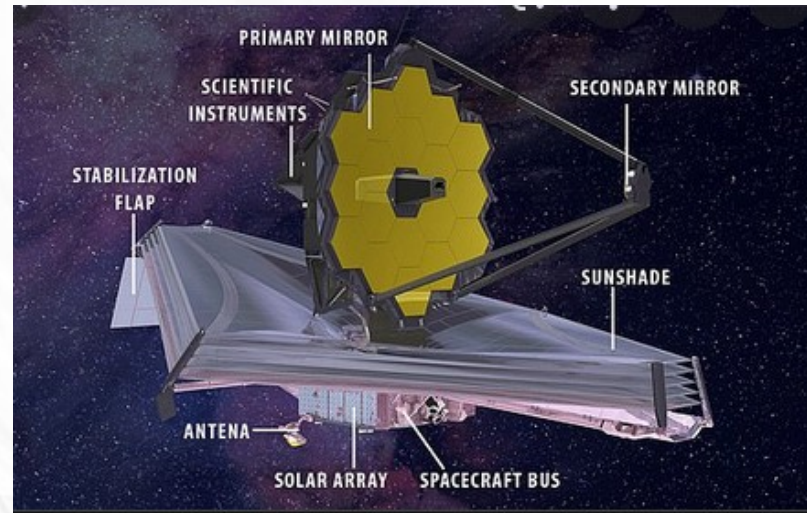
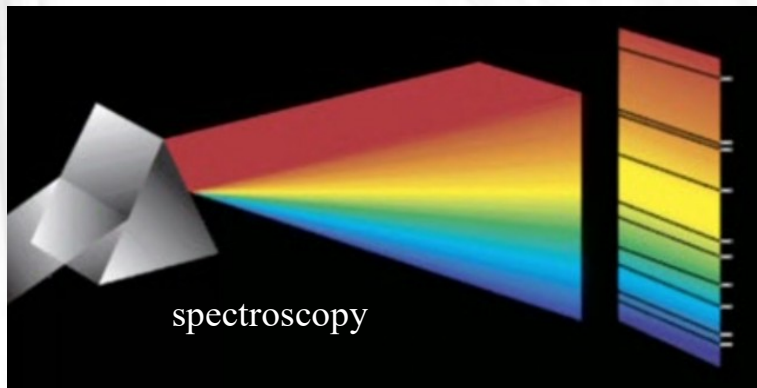


Video
James Webb
Last pics



KAIT telescope

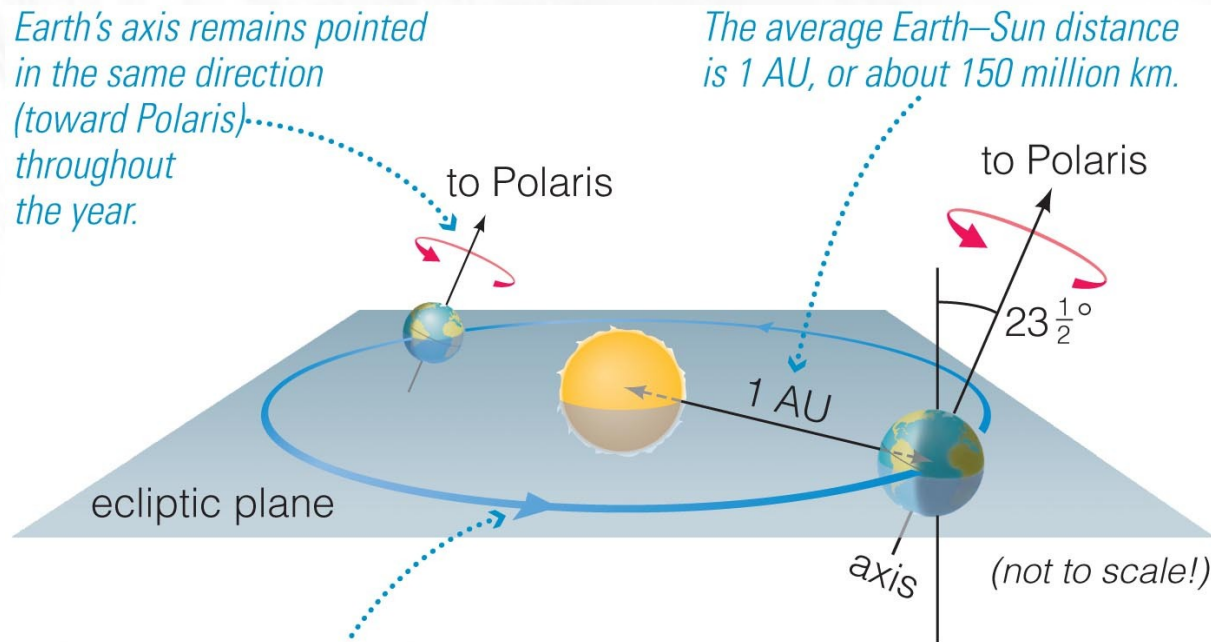
Lick Observatory (Mt Hamilton) , CA



Earth **orbits** the Sun (revolves) once every year:

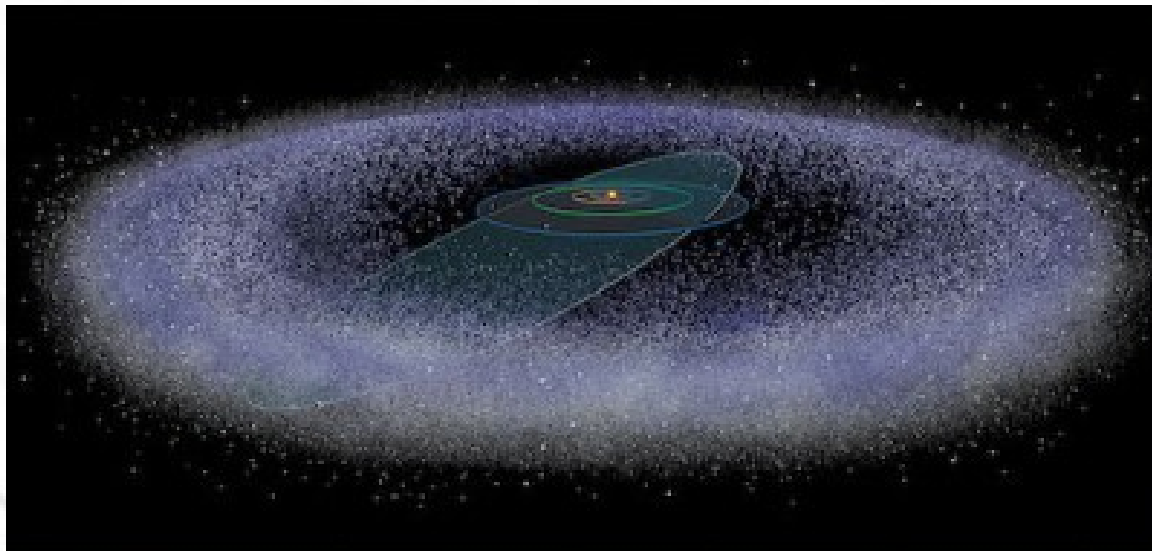
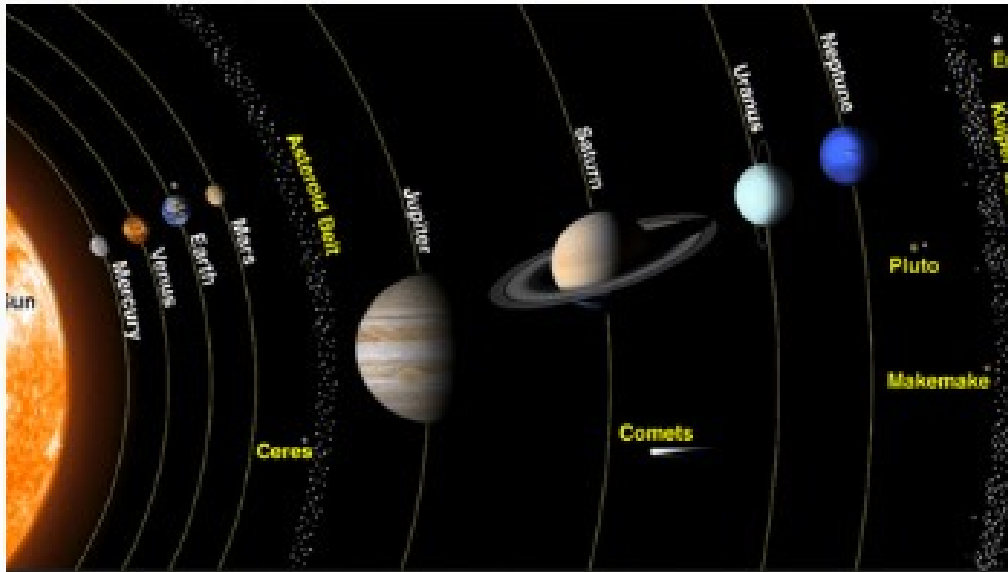
- at an average distance of 1 AU $\approx$ 150 million kilometers.
- with Earth's axis tilted by 23.5° (pointing to Polaris)

It rotates in the same direction it orbits, **counterclockwise** as viewed from above the North Pole.



Earth takes 1 year to orbit the Sun at an average speed of 107,000 km/hr.

Convert to miles per hour



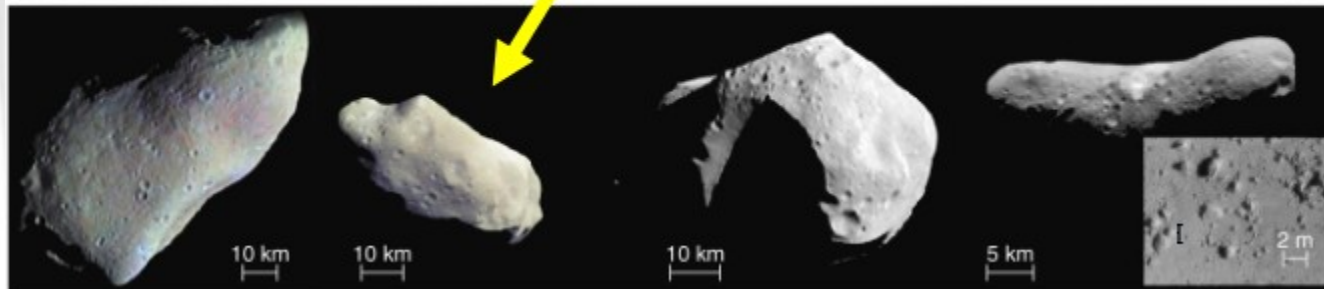
Catastrophic collision – asteroids → video_Jupiter (1994)



Meteor Crater



243 Ida and its moon
Dactyl



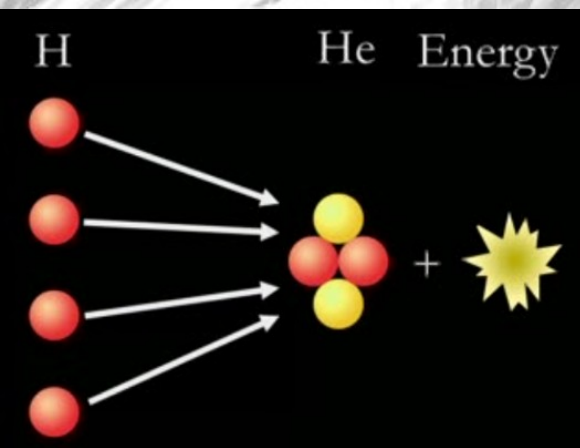
Nebular hypothesis – other planets/



On 9 January 1992, **radio astronomers Aleksander Wolszczan and Dale Frail** announced the discovery of **two planets orbiting the pulsar PSR 1257+12**. This discovery was confirmed, and is generally considered to be the first definitive detection of exoplanets.

Arecibo radio telescope

Video Sun



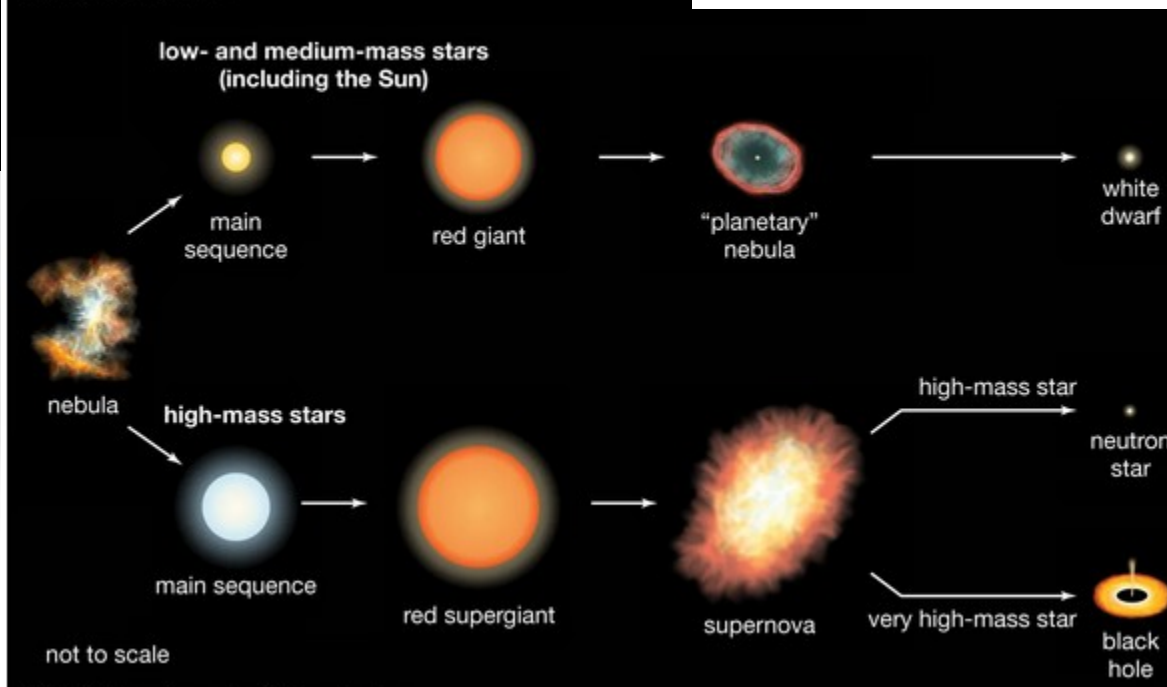
Stellar parameters. Life cycle. Death of stars

Live hard, die young

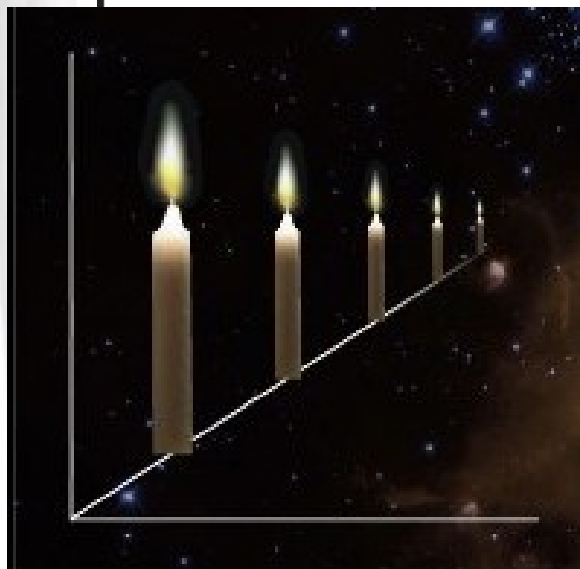
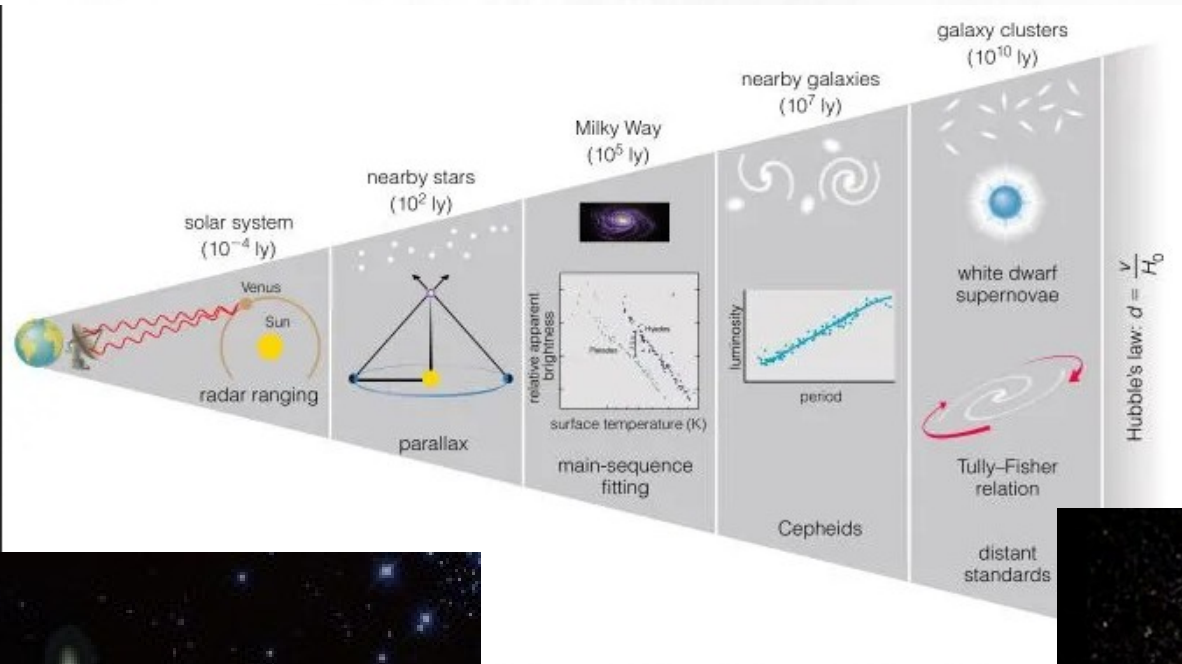
https://www.youtube.com/watch?v=GYYWc_VXJ4A

A star's life expectancy depends on its mass. Generally, **the more massive the star, the faster it burns up its fuel supply, and the shorter its life.** The most massive stars can burn out and explode in a supernova after only a few million years of fusion.

Stellar evolution

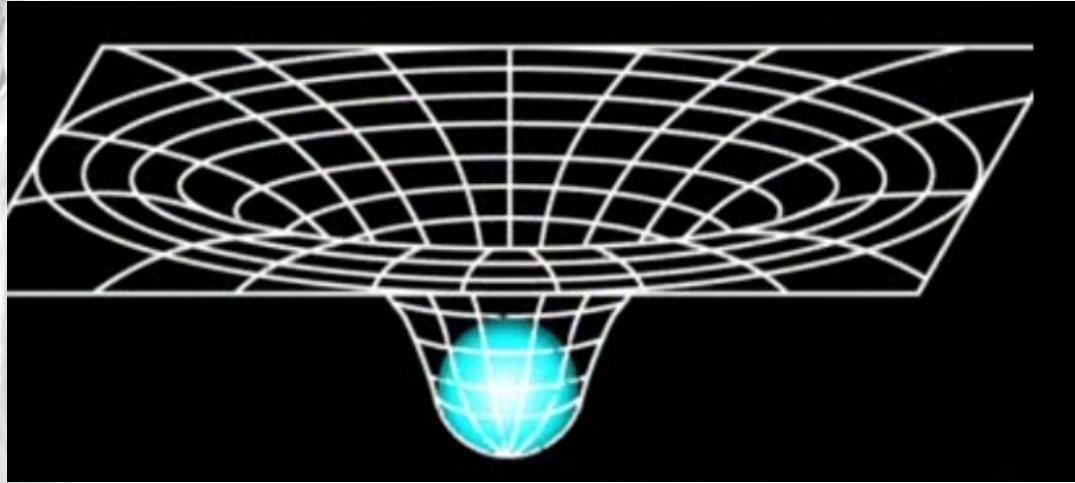


Distance of objects → age of objects by looking at
Clusters of stars.



Oldest object

Stellar corpses . BH. Gravitational waves → video



1,000,000 times denser than Sun
(earth size)

Size of a city = neutron star 10^{14}
denser sun. (1.4 mass)

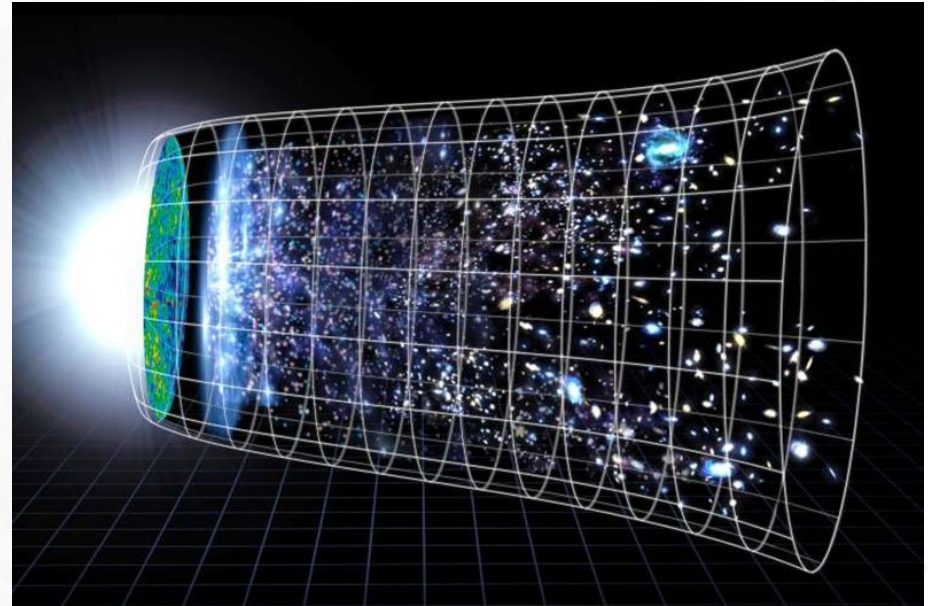
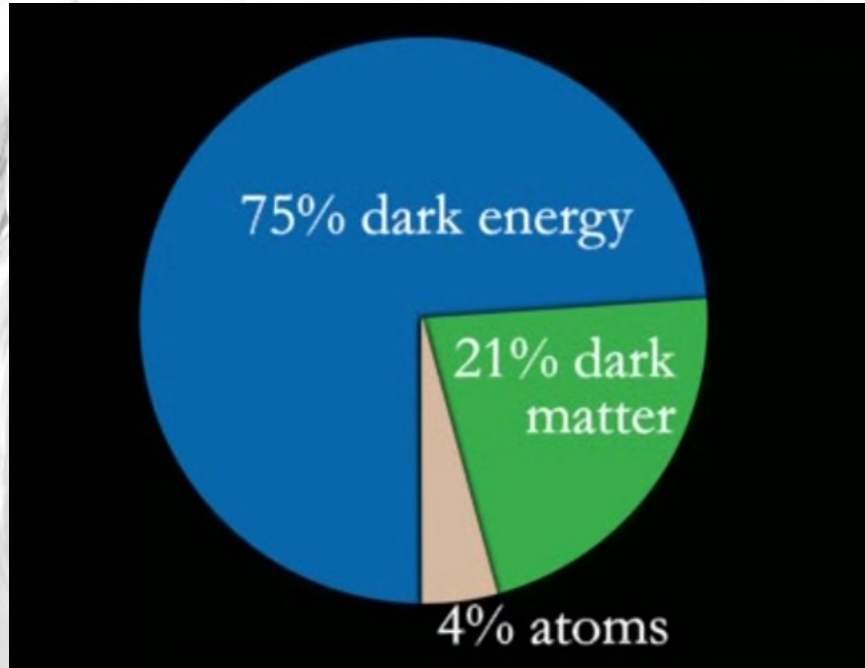


In **2015**, scientists detected gravitational waves for the very first time. They used a very sensitive instrument called LIGO (Laser Interferometer Gravitational-Wave Observatory). These first gravitational waves happened when two black holes crashed into one another.

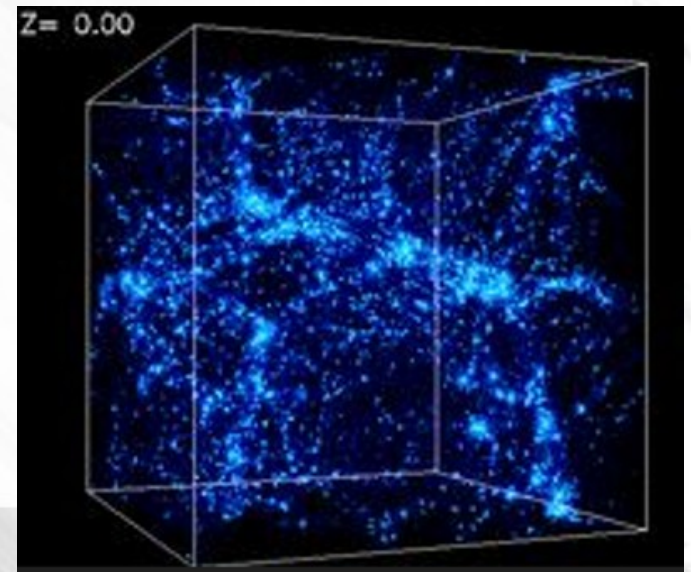
Video gravitational wave



Universe – Expansion – big bang



Video early Universe,
creation of super cluster of galaxies
Ref: Alex Fillipenko .
Understanding the Universe



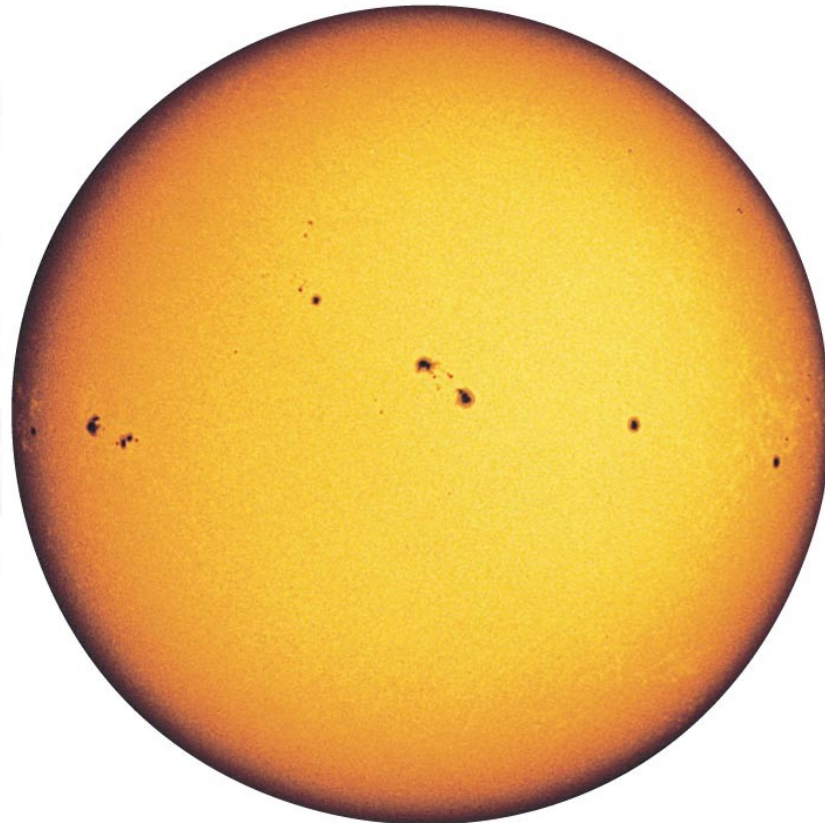
A FEW DEFINITIONS:

BASIC ASTRONOMICAL OBJECTS

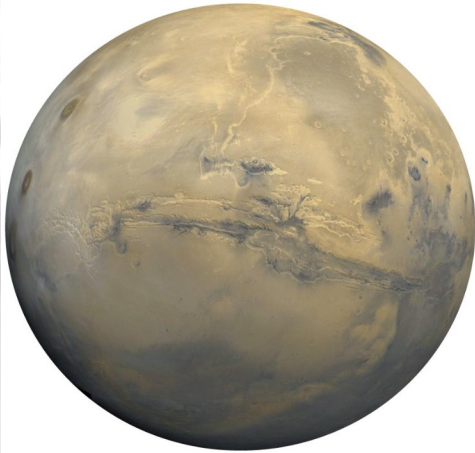


Star

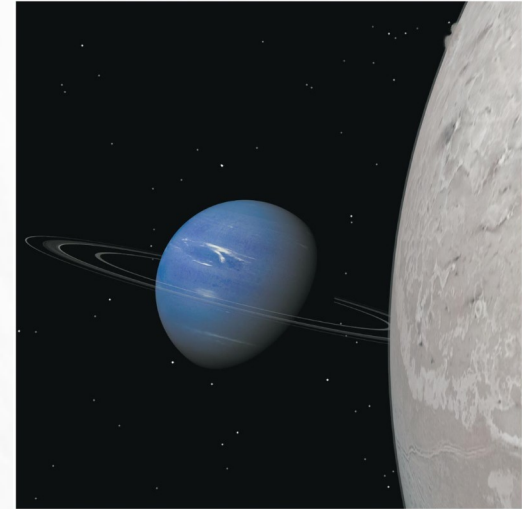
A large, glowing ball of gas that generates heat and light through nuclear fusion



Planet



Mars



Neptune

Shape ? Composition ? Examples ?

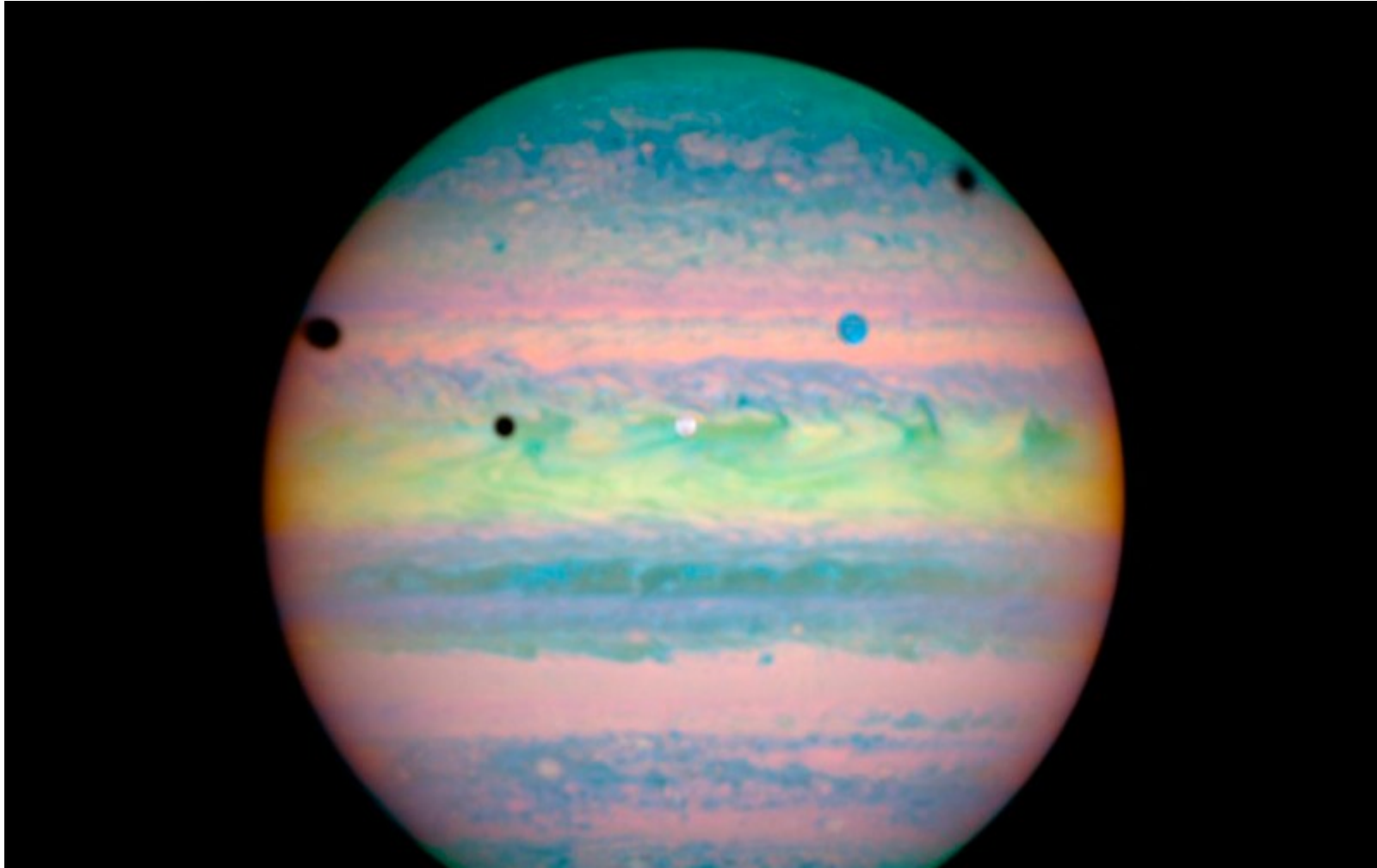
Remember:

My very educated mother just served us nine pizza

Or nothing

A moderately large object that orbits a star; it shines by reflected light. Planets may be rocky, icy, or gaseous in composition.

Which planet is that ? What are the shadows ?



**Do you know your planets ?
What are the first moons of Jupiter ?**

Moon (or Satellite)

An object that orbits a planet

Using newton's version of Kepler's third law, we can use a Moon or Satellite to find the mass of the object being orbited.

$$P^2 = \left[\frac{4\pi^2}{G(m_1 + m_2)} \right] a^3$$

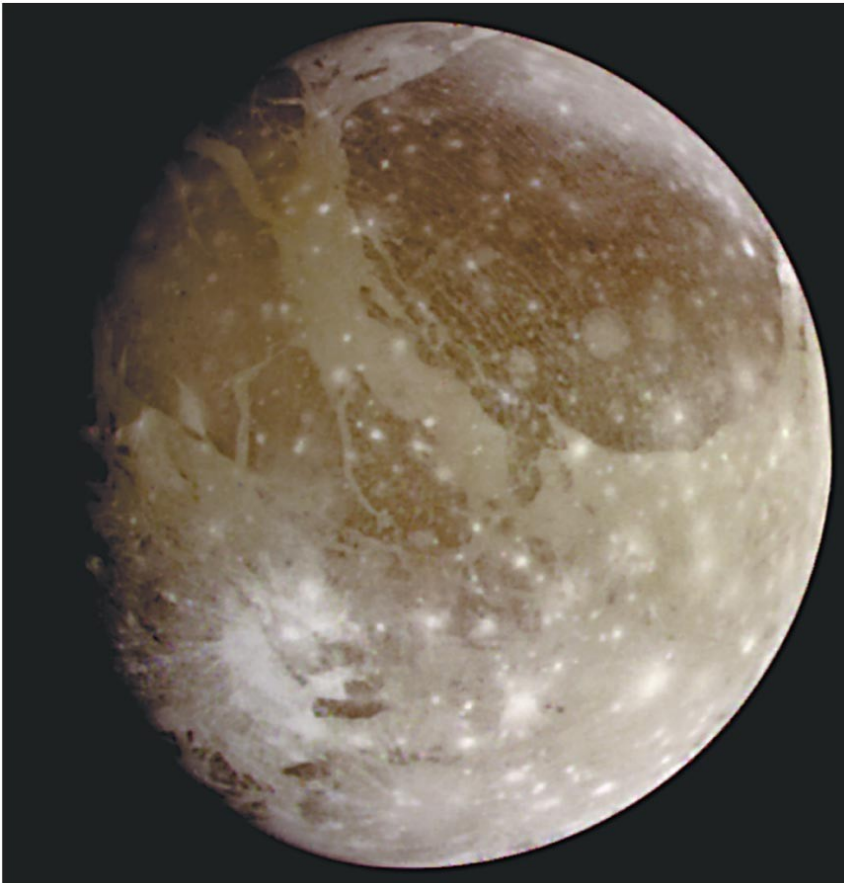
P = sidereal period in seconds

a = semimajor axis in meters

m₁ = mass of first object in kg

m₂ = mass of second object in kg

G = Universal gravity constant



Ganymede (orbits Jupiter)

Comet



A relatively small and icy object that orbits a star

Nebula



An interstellar
cloud of gas
and/or dust

See orion nebula
1500 light years away.

Galaxy

A great island of stars in space, all held together by gravity and orbiting a common center



M31 - the great galaxy

2.5 million light years away | trillion stars

Milky Way Galaxy is estimated to contain roughly 250 billion stars

Universe

The sum total of all matter and energy;
that is, everything within and between all
galaxies

